eLife

Neuroscience

Combinatorial expression of gamma-protocadherins regulates synaptic specificity in the mouse neocortex

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted
August 31, 2023 (this version)

Sent for peer review
May 30, 2023

Posted to bioRxiv
May 23, 2023

Yi-Jun Zhu, Cai-Yun Deng, Liu Fan, Ya-Qian Wang, Hui Zhou, Hua-Tai Xu

Institute of Neuroscience and State Key Laboratory of Neuroscience, CAS Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, 200031, China • Lingang Laboratory, Shanghai Center for Brain Science and Brain-Inspired Intelligence Technology, Shanghai 201210, China • University of Chinese Academy of Sciences, Beijing 100049, China

Abstract

During the process of synaptic formation, neurons require not only certain principles for selecting partners to form synapses but also additional mechanisms to avoid undesired cells. However, the means to get around unwanted partners remains largely unknown. In this study, we have identified that the combinatorial expression of clustered protocadherin gammas (γ-PCDHs) is essential in regulating such specificity in the mouse neocortex. Using single-cell sequencing from the 5-prime end, we revealed the combinatorial expression pattern of γ-PCDH variable isoforms in neocortical neurons. Whole-cell patch-clamp recordings demonstrated that increasing the similarity level of this combinatorial pattern in neurons reduced their synaptic connectivity. Our findings reveal a delicate molecular mechanism for assembling the neural network in the mouse neocortex.

eLife assessment

This study presents a **valuable** finding on the functional significance of protocadherin gamma (PCDHg) isoforms in establishing or maintaining functional synaptic connectivity. The authors used sophisticated methods such as single-cell sequencing and whole-cell patch-clamp recordings and obtained **solid** results that suggest that the neurons expressing the same PCDHg isoforms are less likely to be synaptically connected to each other. These results are largely consistent with previous results using morphological criteria.

Introduction

The specificity of synaptic connections is essential for the function of neural circuits(Yogev and Shen, 2014). Cell adhesion molecules play crucial roles in the specificity of synapse formation(Duan et al., 2014, Serizawa et al., 2006, Tan et al., 2015, Rawson et al., 2017, Sytnyk et al., 2017, Berns et al., 2018, Jang et al., 2017, Courgeon and Desplan, 2019), but how to achieve specificity at the microcircuit level remains to be determined. The unique expression pattern of clustered protocadherins (cPCDH) leads to millions of possible combinations of cPCDH isoforms on the neuron surface(Kaneko et al., 2006, Esumi et al., 2005), suggesting cPCDHs function as a unique barcode for each neuron(Yagi, 2012). The absence of γ-PCDH does not cause general abnormality in the development of the cerebral cortex, such as cell differentiation, migration, and survival(Wang et al., 2002). However, the presence of γ-PCDH has been detected at the synaptic contacts(Fernandez-Monreal et al., 2009, Phillips et al., 2003, LaMassa et al., 2021), and its absence has substantial effects on neuronal connectivity(Tarusawa et al., 2016, Kostadinov and Sanes, 2015, Lv et al., 2022). The homophilic property of γ-PCDH allows them to promote exuberant dendritic complexity(Molumby et al., 2016). However, more evidence suggests that it might be detrimental to synapse formation. Previous studies indicated that homophilic interactions mediated by large overlapping patterns of cPCDH isoforms on opposing cell surfaces might lead to intercellular repulsion(Rubinstein et al., 2015, Brasch et al., 2019, Honig and Shapiro, 2020, Lefebvre et al., 2012), in which the interaction with the neuroligin family may be involved(Molumby et al., 2017, Steffen et al., 2021).

Following this repulsion concept, the absence of γ-PCDH caused significantly more dendritic spines and inhibitory synapse densities in neocortical neurons(Molumby et al., 2017, Steffen et al., 2021). Consistently, neurons with overexpressed one of the γ-PCDH isoforms had significantly fewer dendritic spines from the same condition(Molumby et al., 2017). Furthermore, the absence of the clustered PCDH also increased local reciprocal neural connection between lineage-related neurons in the neocortex(Tarusawa et al., 2016, Lv et al., 2022), even though sister cells had more similar expression patterns of γ-PCDH isoforms(Lv et al., 2022).

Although γ-PCDH has these “contradictory” effects on dendrites’ complexity and dendritic spines, it is detrimental to synapse formation in the forebrain. However, each neuron expresses multiple isoforms of γ-PCDH(Kaneko et al., 2006, Lv et al., 2022). What could be the impact of this combinatorial expression on synapse formation? Here, using 5-prime end (5’-end) single-cell sequencing, we revealed the diversified combinatorial expression of γ-PCDH isoforms in neocortical neurons. With multiple whole-cell patch-clamp recordings after the sequential *in utero* electroporation, we discovered that the combinatorial expression of γ-PCDHs plays a vital role in synaptic specificity to assemble the neural microcircuit in the mouse neocortex. This is achieved by allowing a neuron to choose which partners not to form a synapse with rather than choosing which ones to make synapses with.

Results

The diversified combinatorial expression pattern of γ-PCDHs in neocortical neurons revealed by 5’-end single-cell sequencing

The gamma isoform of cPCDHs (γ-PCDHs) is critical for synaptic connectivity(Kostadinov and Sanes, 2015, Tarusawa et al., 2016, Lv et al., 2022). To determine the role of γ-PCDH in the neocortex, we examined their expression in the neocortical neurons of postnatal mice. Previous studies have suggested that cPCDHs were stochastically expressed in neurons(Hirayama et al., 2012, Toyoda et al., 2014). Here, we applied a 5’-end single-cell RNA sequencing to identify γ-PCDH isoforms by sequencing their variable exon (exon 1), where they differ from each other(Kohmura et al., 1998, Wu and Maniatis, 1999). Since the second postnatal week is the critical stage for synapse formation in the rodent neocortex(Lendvai et al., 2000, Holtmaat and Svoboda, 2009), we chose postnatal day 11 (P11) to examine their expression. After reverse transcription and cDNA amplification (**Fig. 1**-S1A, B), we split cDNA into two parts: one for specific amplification of *pcdhg* mRNAs and the other for the 5’ gene expression library construction (**Fig. 1**-S1C). After the cluster analysis (**Fig. 1**-S2-4), we collected 6505 neurons out of 17438 cells (**Fig. 1A** and **Fig. 1**-S1D). Neurons with more than 10 UMI (cutoff>1 for each type of individual isoform) for all γ-PCDH (**Fig. 1B, C**) were further analyzed. We found that significantly more cells expressed “C-type” isoforms (C3, C4, and C5) compared to other “variable” isoforms (**Fig. 1D** and **Fig. 1**-S1E), which is consistent with a previous study(Toyoda et al., 2014). We performed the pair-wise analysis on similarity between neurons in the single-cell expression pattern of γ-PCDH variable isoforms, which revealed that most neocortical neurons had distinct combinatorial expression patterns (**Fig. 1E**). Based on the variance analysis(Wada et al., 2018) of fraction distribution of the number of the expressed isoforms per cell for all neurons (**Fig. 1F** and **Fig. 1**-S5), we found a weak but significant co-occurrence for the expression of γ-PCDH isoforms in most neurons (**Fig. 1G**). We did not detect any apparent differences among clusters except for cluster 0, which had much less expression (**Fig. 1C**) and no co-occurrence of γ-PCDH isoforms (**Fig. 1G** and **Fig. 1**-S5). Since all of our electrophysiological recordings were carried out on pyramidal neurons in the layer 2/3 of the neocortex, we analyzed the corresponding cluster 7 in more detail. They also showed distinct expression patterns as a general population of neocortical neurons (**Fig. 1**-S6). In summary, the data from 5'-end single-cell sequencing reveal that γ-PCDHs are diversely expressed in most neocortical neurons with a combinatorial pattern.

The absence of γ-PCDHs led to an increase in local synaptic connectivity among pyramidal neurons

To investigate the function of γ-PCDH in the synaptic formation between neocortical neurons, we conducted paired recordings on pyramidal neurons in the layer 2/3 of the neocortex from γ-PCDH conditional knockout (cKO) mice. These mice were obtained by crossing *Pcdhg* $^{flox/flox}$ mice(Lefebvre et al., 2008) with *Nex-cre* mice(Goebbels et al., 2006), in which all variable and C-type γ-PCDH isoforms were specifically removed in pyramidal neurons. Using multiple whole-cell patch-clamp recordings from cortical slices of P9-32 mice, we measured the connectivity among nearby pyramidal cells (<50 μm between cell somas) in the layer 2/3 of the neocortex by the presence of evoked monosynaptic responses (**Fig. 2** and **Fig. 2**-S1). As shown in the sample traces in the connectivity matrix from six recorded neurons (**Fig. 2A** and **Fig. 2**-S1), two neuronal pairs (the neuronal pairs 4 → 3 and 5 → 6) exhibited unidirectional monosynaptic connections (indicated by orange arrows), and one pair (the neuronal pair 1 ↔ 3) with bidirectional connection (indicated by green arrows) out of 15 possibilities. Overall, we found that the percentage of connected pairs was significantly higher in *Pcdhg* cKO mice (20.2%, 103/511) than in wild-type (WT) mice (15.0%, 122/813). Due to their potential different roles in the synaptic organizations in the neocortex(Douglas and Martin, 2004), we performed an extra set of recordings in the layer 2/3 of the neocortex from P10-20 mice to separate neuron pairs along their axes vertical *vs*. horizontal to the pial surface in *Pcdhg* cKO mice. We found a more significant difference in the connectivity for vertically aligned cells (18.3%, 94/515 in *Pcdhg* cKO mice *vs*. 11.2%, 73/651 in WT mice for vertically aligned neuron pairs; 12.2%, 27/221 in *Pcdhg* cKO mice *vs*. 9.5%, 28/294 in WT mice for horizontally aligned neuron pairs) (**Fig. 2C**). We also generated *Pcdha* cKO mice (**Fig. 2**-S2-4) and performed the same set of experiments for vertically aligned neurons. We did not find a significant difference (11.3%, 38/337 in *Pcdha* cKO mice *vs*. 14.3%, 26/182 in WT mice, **Fig. 2D**) in the mice without α-PCDH. In a more detailed analysis of synaptic connections among vertically aligned neurons in *Pcdhg* cKO mice, we found that the absence of γ-PCDH expression appeared to significantly increase synapse formation between cells separated vertically by 50 to 100 μm (24.0%, 50/208 in *Pcdhg* cKO mice *vs*. 9.6%, 20/208 in WT mice) (**Fig. 2E**). Furthermore, we found that the absence of γ-PCDH increased synapse formation starting from P10 (16.1%, 32/199 in *Pcdhg* cKO mice *vs*. 8.7%, 20/230 in WT mice for P10-12; 21.6%, 22/102 in *Pcdhg* cKO mice *vs*. 12.8%, 29/227 in WT mice for P13-15), when chemical synapses between cortical neurons become more detectable, and such tendency maintained for as long as measurements were performed (up to P20) (**Fig. 2F**). Thus, γ-PCDH may play a role in preventing synapse formation starting from the early stage of neural development.

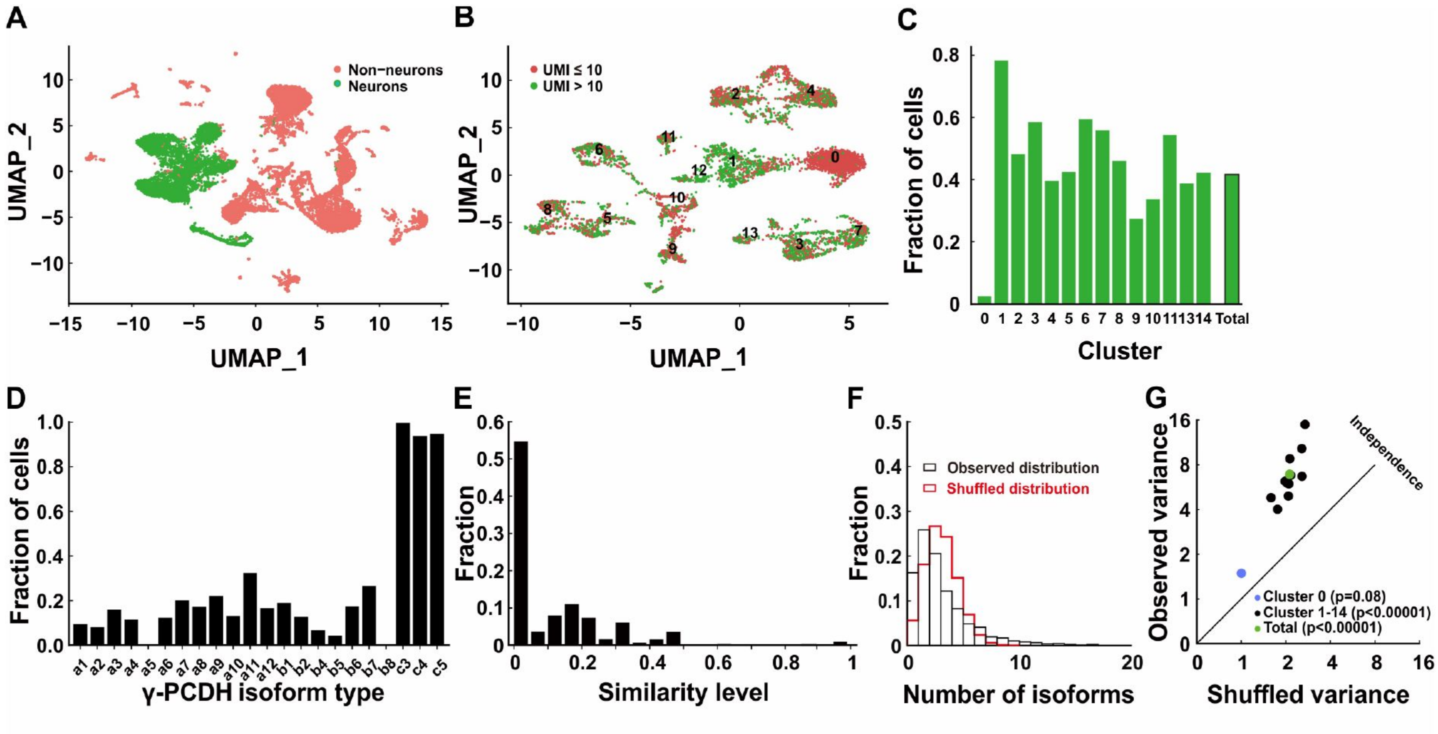

Figure 1

Diversified expression of *pcdhg* isoforms in neocortical neurons.

(**A**) UMAP analysis of all cells (n=17438) collected from 5′-end single-cell sequencing after the cleanup and doublets removing. Neurons were labeled as green dots and non-neurons as red. (**B**) UMAP clusters of all neurons further labeled by the UMI cutoff. Red dots represented cells with no more than 10 total UMIs of *pcdhg* (n=3671), and green dots represented cells with more than 10 UMIs (n=2834). (**C**) Fractions of neurons with more than 10 UMIs in different clusters. (**D**) Fractions of neurons expressing different *pcdhg* isoforms in the neocortex. (**E**) Fraction distribution for different similarity levels in the combinatorial expression of *pcdhg* variable isoforms among neurons. The similarity level was calculated as. $\frac{A\cap B}{A\cup B}$. (**F**) The observed distribution (black) for the fraction of cells with a various number of isoforms for all neurons. The shuffled distribution (red) was generated by the hypothesis that all the isoform expressed stochastically. (**G**) The difference between observed and shuffled variance of the fraction distribution of cells from all clusters.

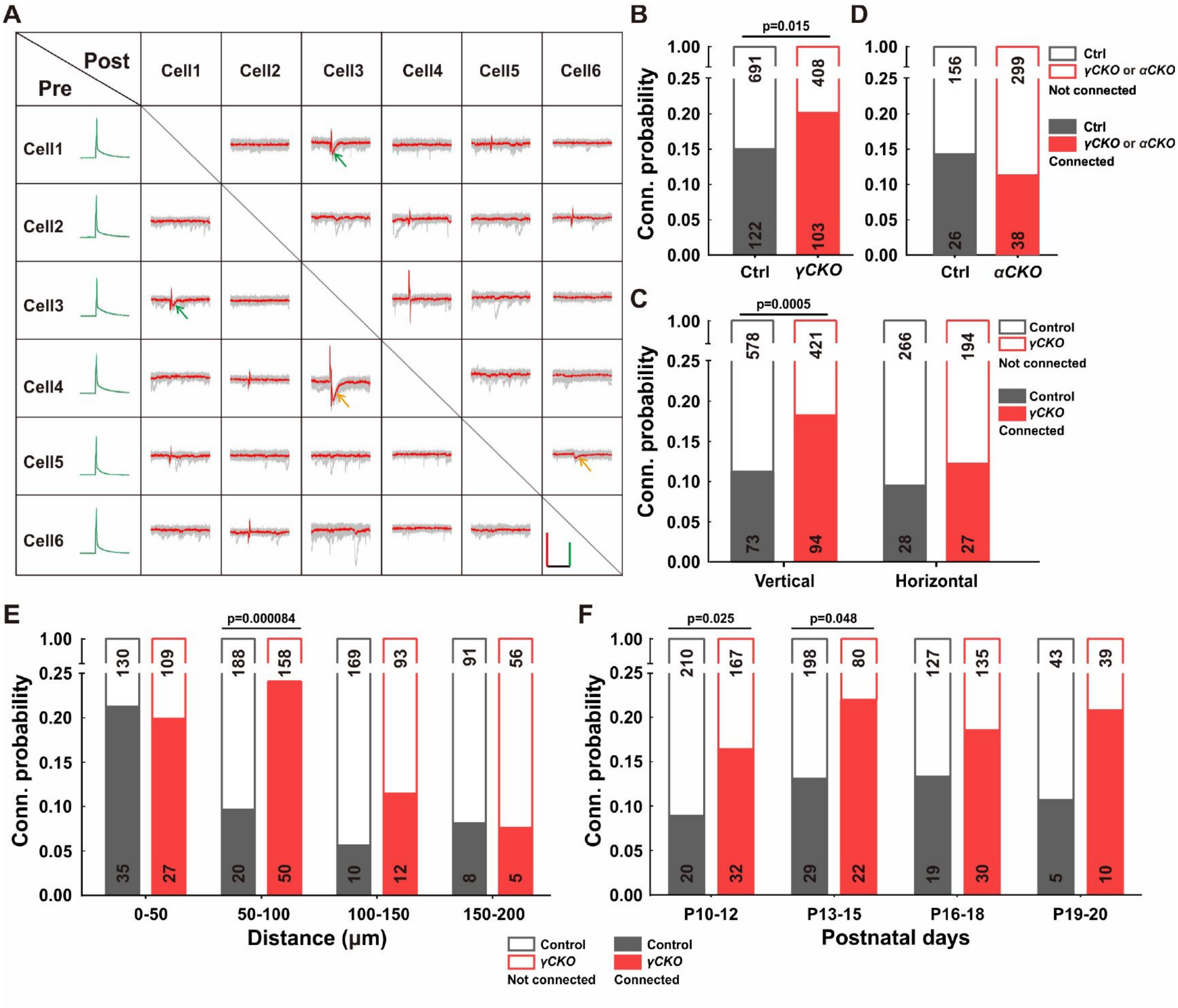

Figure 2

The absence of γ-PCDH increased the synaptic connectivity.

(**A**) Sample traces (red/green, average traces; gray, 10 original traces) of a multiple electrode whole-cell patch-clamp recording on six neurons in the layer 2/3 of the barrel cortex. Positive evoked postsynaptic responses were indicated by arrows. Orange/green arrows: unidirectional/ bidirectional synaptic connections. Scale bars, 100 mV (green), 50 pA (red), and 50 ms (black). (**B**) Connectivity probability among nearby pyramidal cells in the layer 2/3 of the barrel cortex in *pcdhg* cKO mice and their littermate WT controls. γKO: *pcdhg flox/flox*::*nex-cre* mice; Ctrl: *pcdhg*+/+::*nex-cre*. (**C**) Connectivity probability among vertically or horizontally aligned neurons in the layer 2/3 of the barrel cortex in *pcdhg* cKO mice and their littermate WT controls. (**D**) Connectivity probability among vertically aligned pyramidal cells in the layer 2/3 of the barrel cortex in *pcdha* cKO mice and their littermate WT controls. αKO: *pcdha flox/flox*::*nex-cre* mice; Ctrl: *pcdha*+/+::*nex-cre* mice. (**E**) Connectivity probability among vertically aligned pyramidal cells along with the distance between recorded pairs in *pcdhg* cKO mice or WT mice. (**F**) Developmental profiling of the connectivity probability among vertically aligned neurons in *pcdhg* cKO mice or WT mice. Chi-square tests were used in **B**-**F** to calculate the statistical difference.

Overexpression of γ-PCDHs decreases local synaptic connectivity in the mouse neocortex

To further determine the effect of γ-PCDHs on synapse formation, we overexpressed randomly selected single or multiple γ-PCDH isoforms tagged with fluorescent proteins through *in utero* electroporation in mice. We then performed multiple whole-cell patch-clamp recordings to address the effect of γ-PCDHs on pair-wise synaptic connectivity for neurons in the layer 2/3 of the neocortex.

To confirm the overexpression effect, we performed the assays of RT-qPCR (**Fig. 3A, B**), single-cell RT-PCR (**Fig. 3C, D**), and immunohistochemistry (**Fig. 3**-S1A to C). The RT-qPCR assay revealed that the electroporated isoforms, but not the non-overexpressed ones, from the surgical side indeed had significantly higher expression levels than those from the contralateral side (**Fig. 3A, B**). To estimate how many isoforms were expressed in a given neuron when multiple plasmids were electroporated, we tagged first five isoforms with mNeongreen and the sixth one with mCherry (**Fig. 3**-S1A). Based on the rates of yellow and red-only cells in the total electroporated neuron population (**Fig. 3**-S1B), the probability analysis revealed that each positive neuron expressed an average of 5.6 types of isoforms (**Fig. 3**-S1C and the top panel of D). This result was consistent with the data from single-cell RT-PCR analysis in which an average of 5.3 types out of 6 electroporated isoforms was detected from 19 neurons (**Fig. 3C, D** and the bottom panel of **Fig. 3**-S1D). The results from single-cell RT-PCR also suggest that electroporation-introduced isoforms are the dominant ones in these neurons (**Fig. 3D**).

Since the layer distribution of neocortical neurons plays a critical role in their synaptic connection(He et al., 2015), we examined their positions relatively to the pial surface after the overexpression. We found that overexpressing γ-PCDH isoforms did not affect cell positions in the neocortex compared to the control plasmids (**Fig. 3**-S1E, F).

We then used multiple recordings to examine the impact of γ-PCDHs on synapse formation. Overexpressing of one or six γ-PCDH variable isoforms in neurons significantly reduced the synaptic connection rate among them (10.3%, 15/146 in expressing control plasmids; 1.9%, 4/216 in overexpressing one isoform; and 4.4%, 8/181 in overexpressing six isoforms, red bars in **Fig. 3E**). However, overexpressing γ-PCDH C4 did not affect the synaptic connection rate (11.3%, 12/106 in overexpressing γ-PCDH C4, **Fig. 3E**). To exclude further the potential effect of C-types of γ-PCDHs' impacts in synapse formation, we electroporated six variable isoforms in *pcdhg* cKO mice. The overexpression also led to the reduction of the connection rate in *pcdhg* cKO mice (6.1%, 12/198 in overexpressing six isoforms vs. 16.5%, 31/188 in expressing control plasmids), similar to what we observed in WT littermate mice (6.6%, 8/121 in overexpressing six isoforms vs. 16.5%, 18/109 in expressing control plasmids) (**Fig. 3F**). These observations reveal that overexpressing the variables, but not the C-types of γ-PCDHs isoforms, decrease synaptic connectivity in the mouse neocortex.

Combinatorial expression of γ-PCDHs regulates synaptic specificity in the mouse neocortex

To investigate the impact of the combinatorial expression pattern of γ-PCDH isoforms on synapse formation, we conducted a sequential *in utero* electroporation at E14.5 and E15.5 (**Fig. 4A**). By electroporating the same or different combinations of γ-PCDH variable isoforms on these two days, we aimed to manipulate the expression similarity level between two groups of neurons. Isoforms were randomly selected based on their comparable expression, except for isoforms A5 and B8, which had low expression levels according to the single cell sequencing data (**Fig. 1D** and **Fig. 1**-S1E). We assigned their combinations to achieve similarity levels ranging from 0% (no overlap, completely different) to 100% (complete overlap, all identical) (**Fig. 4B**). We used fluorescent proteins mNeongreen and mRuby3(Bajar et al., 2016) to tag isoforms for

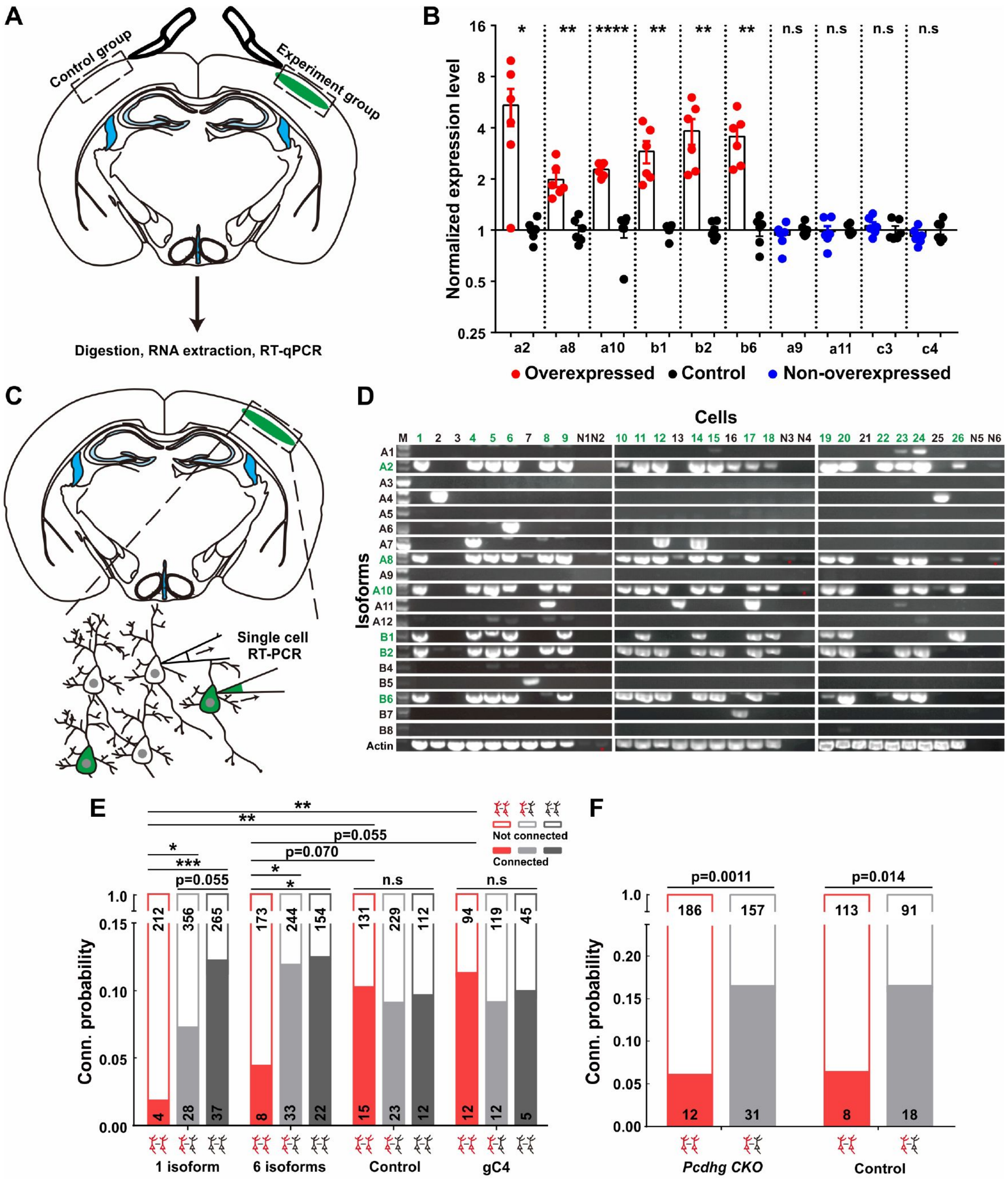

Figure 3

Overexpressing identical variables, but not C-types γ-PCDHs in neurons decreased their synaptic connectivity.

(**A**) The sketch to show the brain regions used for RT-qPCR for both experimental and control groups. (**B**) The overexpression levels were tested by RT-qPCR of electroporated regions. Electroporated isoforms: red; control isoforms: blue. The contralateral sides were used as controls shown in black. Student's *t test* was used, *: p<0.05, **: p<0.01, ***: p<0.001, ****: p<0.0001. (**C**) The diagram of extracting cells for the single-cell RT-PCR assay. (**D**) Single-cell RT-PCR for γ-PCDH isoforms after the electroporation. Green numbers indicated neurons with fluorescence, and black ones represented nearby neurons without fluorescence. N1-N6 were negative controls for the PCR. Isoforms used in the electroporation were shown in green. Red stars indicate faint signals in negative controls. (**E**) The effect of overexpressing one or six γ-PCDH isoforms on the synaptic connectivity in WT mice. 1 isoform: *pcdhga2*; 6 isoforms: *pcdhga2, pcdhga8, pcdhga10, pcdhgb1, pcdhgb2*, and *pcdhgb6*; gC4: *pcdhgC4*; Control: the plasmid vector without *pcdhg* insertion. (**F**) The effect of overexpressing 6 γ-PCDH isoforms on the synaptic connectivity in *pcdhg* cKO mice. 6 used isoforms were same with the ones in (**E**). *pcdhg cKO*: *pcdhg* conditional knockout mice; Control: WT littermates. Chi-square test and false discovery rate (FDR) correction were used to determine the statistical differences between groups in (**E**) and (**F**).

electroporation at E14.5 and E15.5 respectively (**Fig. 4A**). Subsequently, whole-cell patch-clamp recordings were conducted on layer 2/3 neurons in the neocortex from acute brain slices containing both green and red cells from P10-14 pups. Each set of recordings included at least one mNeongreen$^+$, one mRuby3$^+$, and one nearby control neuron without fluorescence (**Fig. 4C**). Our results showed that neurons with the same color exhibited significantly lower connectivity (**Fig. 4**-S1A), which aligns with the findings from single electroporation (as shown in **Fig. 3E**). In the complete overlap (100%) group, the connectivity rate between neurons electroporated at different days was also significantly lower compared to the control (3.7%, 5/134 in the complete-overlap group; 12.5%, 19/151 in control pairs, 100% in **Fig. 4D**). However, as the similarity levels decreased from 100% to 0%, the connectivity probabilities progressively returned to the control level. The likelihood recovered to 8.4% (14/165) for the pairs with a 33% similarity level and 10.3% (12/117) and 10.5% (12/114) for pairs with 11% or 0% similarity level, respectively (**Fig. 4D**). These observations suggested that it is the similarity level of γ-PCDH isoforms between neurons, but not the absolute expression of the protein in individual neurons, that regulate the synaptic formation. Similar to the single electroporation experiment (gray bars in **Fig. 3E**), there were no significant changes in the synaptic connectivity between electroporated and nearby control neurons (**Fig. 4**-S1B). Overall, our findings demonstrate a negative correlation between the probability of forming synaptic connections and the similarity level of γ-PCDH isoforms expressed in neuron pairs (**Fig. 4E**). These findings strongly suggest that the diversified combinatorial expression of γ-PCDH isoforms is crucial for synapse formation between nearby pyramidal cells. The more similar the γ-PCDH isoforms patterns expressed in neurons, the lower the probability of forming synapses between them (**Fig. 4F**).

Discussion

Homophilic proteins cPCDHs are strong candidates for promoting synaptic specificity due to their combinatorial and stochastic expression pattern(Yagi, 2012, Kohmura et al., 1998, Toyoda et al., 2014). Our 5'-end single-cell sequencing data provided insights into the combinatorial expression pattern of γ-PCDH isoforms in neocortical neurons. We further demonstrated the critical role of this diversity in synaptic specificity through three lines of evidence. Firstly, the absence of γ-PCDH significantly increased functional connectivity between adjacent neocortical neurons. Secondly, electroporation-induced overexpression of identical γ-PCDH variable isoforms in developing neurons markedly decreased their connectivity. Lastly, using sequential *in utero* electroporation with different batchs of isoforms, we found that increasing the similarity level of γ-PCDH variable isoforms expressed in neurons led to a reduction in their synaptic connectivity. These findings suggest that γ-PCDHs regulate the specificity of synapse formation by preventing synapse formation with specific cells, rather than by selectively choosing particular targets. It remains to be studied whether the diversified patterns of γ-PCDH isoforms expressed in different neurons have additional coding functions for neurons beyond their homophilic interaction.

Previous studies by Molumby et al. demonstrated that neurons from the neocortex of *pcdhg* knockout mice exhibited significantly more dendritic spines, while neurons overexpressing a single γ-PCDH isoforms had fewer dendritic spines in (Molumby et al., 2017). Our recordings are consistent with these previous morphology studies. Tarusawa et al. revealed that the absence of the whole cluster of cPCDH affected synaptic connections among lineage-related cells(Tarusawa et al., 2016). More recently, overexpression of the C-type γ-PCDH isoform C3 also showed a negative effect on synapse formation within a defined clone(Lv et al., 2022). In our study, we further demonstrated an increased synaptic connection rate between adjacent pyramidal neurons in the neocortex of *pcdhg* knockout mice, while it decreased between neurons overexpressing single or multiple identical γ-PCDH variable isoforms. These effects were not just limited to lineage-dependent cells. Together with previous findings(Molumby et al., 2017, Tarusawa et al., 2016), our observations solidify the repulsion effect of γ-PCDH on synapse formation among neocortical neurons. Some subtle differences exist between our findings and previous

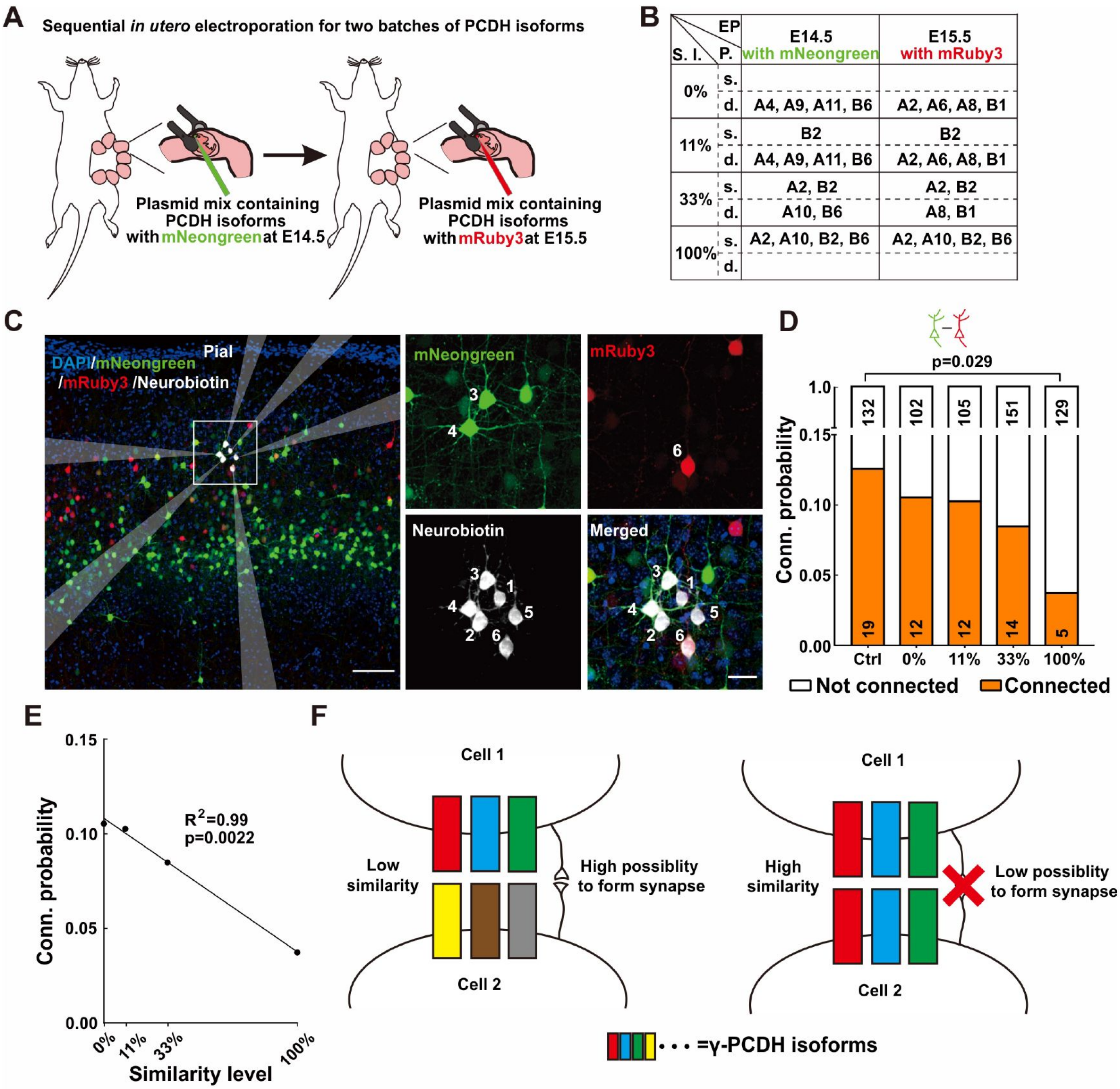

Figure 4

Diversified γ-PCDHs were critical for synapse formation in cortical neurons.

(**A**) The diagram of a sequential *in utero* electroporation at E14.5 and E15.5. (**B**) The overexpressed γ-PCDH isoforms in different sets of experiments to achieve different similarity levels between neurons. S.l., Similarity level; EP, Electroporation; P., plasmids mixture; s./d., same/different isoforms in two electroporations. (**C**) A sample image of recorded neurons after two times of electroporation at E14.5 with plasmids carrying mNeongreen and E15.5 with plasmids carrying mRuby3. In this sample, cells 3 and 4 are mNeongreen positive, cell 6 is mRuby3 positive, and cells 1, 2, and 5 are negative ones without fluorescence. Neurobiotin was added into the internal solution to indicate recorded neurons. The translucent arrows showed the positions of electrodes. Scale bar, left, 100μm, right, 25μm. (**D**) The connectivity probability for neuron pairs overexpressed different batch of γ-PCDHs isoforms (labeled with different fluorescence) after a sequential *in utero* electroporation. Chi-square test and false discovery rate (FDR) correction were used to determine the statistical difference. (**E**) The correlation between the similarity level of overexpressed γ-PCDHs combination and the probability of synaptic connection rate (The source data were the same as d). (**F**) The graph summary of γ-PCDHs' effect on synapse formation.

recordings(Tarusawa et al., 2016, Lv et al., 2022). Tarusawa et al. demonstrated that the connection probability between excitatory neurons lacking the entire cPCDH cluster in the layer 4 was approximately twofold higher at early stage P9-11, significantly lower at P13-16, and similar to control cells at P18-20 compared (Tarusawa et al., 2016). In our study, *pcdhg*$^{-/-}$ pyramidal neuronal pairs consistently exhibited a higher connection probability from P10 to P20. Two potential reasons could explain these differences. Firstly, we only removed γ-PCDH instead of the entire cPCDH cluster, which includes α, β, and γ isoforms. Secondly, γ-PCDH might have different functions in neurons located in the layer 2/3 compared to the layer 4. Lv et al. found that overexpression of C-type γ-PCDH C3 decreased the preferential connection between sister cells(Lv et al., 2022). However, our study demonstrated that only variable, but not C-type isoforms had a negative impact on synapse formation. This discrepancy might be attributed to the lineage relationship, which could have an unknown impact on synapse formation. Since a single neuron can express multiple isoforms, deleting all γ-PCDH isoforms might mask the role of this combination. In this study, we manipulated the combinatorial expression patterns of γ-PCDH isoforms in nearby neocortical neurons through sequential *in utero* electroporation, expressing different batches of isoforms with adjustable similarities. We observed that when two neurons expressed identical variable isoforms (100% group), the likelihood of synapse formation between them was lowest. As the similarity level between two cells decreased, with fewer shared isoforms, the connectivity probability increased. The connectivity probability between neurons with different variable isoforms (0% group) did not differ from the control pairs (without overexpression). However, the connections between overexpressed and control neurons were not affected under both 100% and 0% similarity conditions. These observations suggest that the similarity level, rather than the absolute expression of the protein, affects synapse formation between neurons.

Our findings also demonstrated that the overexpression of multiple γ-PCDH variable isoforms in one neuron only affected its connection if the other neuron overexpressed an identical combination of PCDHg isoforms. This indicates that the expression of multiple PCDHg variable isoforms in a neuron assists in selecting its synapse partner, emphasizing the essential role of diversified combinatorial expression of γ-PCDHs in synaptic specificity in the mouse neocortex. Moreover, the overexpression of the γ-PCDH C4 isoform did not affect synaptic connections, while overexpression of six variable isoforms led to a reduction in the connection rate in *pcdhg* cKO mice. These findings suggest that the variables, but not the C-type isoforms of γ-PCDHs, play crucial roles in the synapse formation in the mouse neocortex.

While the absence of γ-PCDHs causes significantly more synaptic formation among neocortical pyramidal neurons, evidence supports that their absence also leads to a significant reduction of synapse formation in other brain regions. For example, mice lacking γ-PCDHs exhibit fewer synapses in spinal cord interneurons(Weiner et al., 2005). Knocking down γ-PCDHs causes a decline in dendritic spines in cultured hippocampal neurons(Suo et al., 2012) and decrease the astrocyte-neuron contacts in the coculture from the developing spinal cord(Garrett and Weiner, 2009). The absence of γ-PCDHs leads to the reduction of dendritic arborization and dendritic spines in olfactory granule cells(Ledderose et al., 2013). Additionally, immuno-positive signals for γ-PCDHs are more frequently detected in mushroom spines than in thin spines(LaMassa et al., 2021). Moreover, we observed that the absence of γ-PCDHs impacted vertical-aligned neurons more than horizontal-situated pairs in the neocortex. These findings suggest that different mechanisms may be employed by synapses in different brain regions to achieve their specificity. Notably, different mechanisms have already been proposed for targeting specific inhibitory neural circuits in the neocortex, including “on-target” synapse formation for targeting apical dendrites and “off-target” synapse selective removal for somatic innervations(Gour et al., 2021).

In summary, our data demonstrate that the similarity level of γ-PCDH isoforms between neocortical neurons is critical for their synapse formation, with neurons expressing more similar γ-PCDH isoforms patterns exhibiting a lower probability of forming synapses between them. This suggests that the presence of γ-PCDHs enables neocortical neurons to choose which neurons to avoid synapsing with, rather than selecting specific neurons to form synapses with. Whether there are specific attractive forces between cells to promote synaptic specificity remains an open question.

Acknowledgements

We thank Dr. Yifeng Zhang for providing the *Pcdhg flox/flox* mouse line and Dr. Zilong Qiu for *Nex-cre* mouse line, Dr. Jun Chu for sharing plasmids with mNeongreen and mRuby3, and other members in Xu lab for their discussions and technique supports. We are grateful to Prof. Mu-ming Poo and Song-hai Shi for critical reading of the manuscript. This work was supported by grants from the Training Program of the Major Research Plan of the National Natural Science Foundation of China, grant No. 91632101; Strategic Priority Research Program of the Chinese Academy of Sciences, grant No. XDB32010100; National Natural Science Foundation of China project 31671113; Shanghai Municipal Science and Technology Major Project,grant No. 2018SHZDZX05,the State Key Laboratory of Neuroscience and the Lingang Laboratory, grant No. LG-GG-202201-01.

Author contributions

H. -T. X. conceived the project. Y. -J. Z. performed most recordings and single-cell analysis. C. -Y. D. and L. F. performed part of recordings. Y. -Q. W. prepared PCDH plasmids. H. Z. prepared mice. Y. -J. Z. and H. -T. X. wrote the manuscript.

References

Bajar B. T., Wang E. S., Lam A. J., Kim B. B., Jacobs C. L., Howe E. S., Davidson M. W., Lin M. Z., Chu J. (2016) **Improving brightness and photostability of green and red fluorescent proteins for live cell imaging and FRET reporting** *Sci Rep* **6** https://doi.org/10.1038/srep20889

Berns D. S., Denardo L. A., Pederick D. T., Luo L. (2018) **Teneurin-3 controls topographic circuit assembly in the hippocampus** *Nature* **554**:328–333 https://doi.org/10.1038/nature25463

Brasch J., Goodman K. M., Noble A. J., Rapp M., Mannepalli S., Bahna F., Dandey V. P., Bepler T., Berger B., Maniatis T., Potter C. S., Carragher B., Honig B., Shapiro L. (2019) **Visualization of clustered protocadherin neuronal self-recognition complexes** *Nature* **569**:280–283 https://doi.org/10.1038/s41586-019-1089-3

Courgeon M., Desplan C. (2019) **Coordination between stochastic and deterministic specification in the Drosophila visual system** *Science* **366** https://doi.org/10.1126/science.aay6727

Douglas R. J., Martin K. A. (2004) **Neuronal circuits of the neocortex** *Annu Rev Neurosci* **27**:419–51 https://doi.org/10.1146/annurev.neuro.27.070203.144152

Duan X., Krishnaswamy A., Huerta DE LA, Sanes J. R. (2014) **Type II cadherins guide assembly of a direction-selective retinal circuit** *Cell* **158**:793–807 https://doi.org/10.1016/j.cell.2014.06.047

Esumi S., Kakazu N., Taguchi Y., Hirayama T., Sasaki A., Hirabayashi T., Koide T., Kitsukawa T., Hamada S., Yagi T. (2005) **Monoallelic yet combinatorial expression of variable exons of the protocadherin-alpha gene cluster in single neurons** *Nat Genet* **37**:171–6 https://doi.org/10.1038/ng1500

Fernandez-Monreal M., Kang S., Phillips G. R. (2009) **Gamma-protocadherin homophilic interaction and intracellular trafficking is controlled by the cytoplasmic domain in neurons** *Mol Cell Neurosci* **40**:344–53 https://doi.org/10.1016/j.mcn.2008.12.002

Garrett A. M., Weiner J. A. (2009) **Control of CNS synapse development by {gamma}-protocadherin-mediated astrocyte-neuron contact** *J Neurosci* **29**:11723–31 https://doi.org/10.1523/JNEUROSCI.2818-09.2009

Goebbels S., Bormuth I., Bode U., Hermanson O., Schwab M. H., Nave K. A. (2006) **Genetic targeting of principal neurons in neocortex and hippocampus of NEX-Cre mice** *Genesis* **44**:611–21 https://doi.org/10.1002/dvg.20256

Gour A., Boergens K. M., Heike N., Hua Y., Laserstein P., Song K., Helmstaedter M. (2021) **Postnatal connectomic development of inhibition in mouse barrel cortex** *Science* **371** https://doi.org/10.1126/science.abb4534

He S., Li Z., Ge S., Yu Y. C., Shi S. H. (2015) **Inside-Out Radial Migration Facilitates Lineage-Dependent Neocortical Microcircuit Assembly** *Neuron* **86**:1159–66 https://doi.org/10.1016/j.neuron.2015.05.002

Hirayama T., Tarusawa E., Yoshimura Y., Galjart N., Yagi T. (2012) **CTCF is required for neural development and stochastic expression of clustered Pcdh genes in neurons** *Cell Rep* **2**:345–57 https://doi.org/10.1016/j.celrep.2012.06.014

Holtmaat A., Svoboda K. (2009) **Experience-dependent structural synaptic plasticity in the mammalian brain** *Nat Rev Neurosci* **10**:647–58 https://doi.org/10.1038/nrn2699

Honig B., Shapiro L. (2020) **Adhesion Protein Structure, Molecular Affinities, and Principles of Cell-Cell Recognition** *Cell* **181**:520–535 https://doi.org/10.1016/j.cell.2020.04.010

Jang S., Lee H., Kim E. (2017) **Synaptic adhesion molecules and excitatory synaptic transmission** *Curr Opin Neurobiol* **45**:45–50 https://doi.org/10.1016/j.conb.2017.03.005

Kaneko R., Kato H., Kawamura Y., Esumi S., Hirayama T., Hirabayashi T., Yagi T. (2006) **Allelic gene regulation of Pcdh-alpha and Pcdh-gamma clusters involving both monoallelic and biallelic expression in single Purkinje cells** *J Biol Chem* **281**:30551–60 https://doi.org/10.1074/jbc.M605677200

Kohmura N., Senzaki K., Hamada S., Kai N., Yasuda R., Watanabe M., Ishii H., Yasuda M., Mishina M., Yagi T. (1998) **Diversity revealed by a novel family of cadherins expressed in neurons at a synaptic complex** *Neuron* **20**:1137–51 https://doi.org/10.1016/s0896-6273(00)80495-x

Kostadinov D., Sanes J. R. (2015) **Protocadherin-dependent dendritic self-avoidance regulates neural connectivity and circuit function** *Elife* **4** https://doi.org/10.7554/eLife.08964

Lamassa N., Sverdlov H., Mambetalieva A., Shapiro S., Bucaro M., Fernandez-Monreal M., Phillips G. R. (2021) **Gamma-protocadherin localization at the synapse is associated with parameters of synaptic maturation** *J Comp Neurol* **529**:2407–2417 https://doi.org/10.1002/cne.25102

Ledderose J., Dieter S., Schwarz M. K. (2013) **Maturation of postnatally generated olfactory bulb granule cells depends on functional gamma-protocadherin expression** *Sci Rep* **3** https://doi.org/10.1038/srep01514

Lefebvre J. L., Kostadinov D., Chen W. V., Maniatis T., Sanes J. R. (2012) **Protocadherins mediate dendritic self-avoidance in the mammalian nervous system** *Nature* **488**:517–21 https://doi.org/10.1038/nature11305

Lefebvre J. L., Zhang Y., Meister M., Wang X., Sanes J. R. (2008) **gamma-Protocadherins regulate neuronal survival but are dispensable for circuit formation in retina** *Development* **135**:4141–51 https://doi.org/10.1242/dev.027912

Lendvai B., Stern E. A., Chen B., Svoboda K. (2000) **Experience-dependent plasticity of dendritic spines in the developing rat barrel cortex in vivo** *Nature* **404**:876–81 https://doi.org/10.1038/35009107

Lv X., Li S., Li J., Yu X. Y., Ge X., Li B., Hu S., Lin Y., Zhang S., Yang J., Zhang X., Yan J., Joyner A. L., Shi H., Wu Q., Shi S. H. (2022) **Patterned cPCDH expression regulates the fine organization of the neocortex** *Nature* **612**:503–511 https://doi.org/10.1038/s41586-022-05495-2

Molumby M. J., Anderson R. M., Newbold D. J., Koblesky N. K., Garrett A. M., Schreiner D., Radley J. J., Weiner J. A. (2017) **gamma-Protocadherins Interact with Neuroligin-1 and Negatively Regulate Dendritic Spine Morphogenesis** *Cell Rep* **18**:2702–2714 https://doi.org/10.1016/j.celrep.2017.02.060

Molumby M. J., Keeler A. B., Weiner J. A. (2016) **Homophilic Protocadherin Cell-Cell Interactions Promote Dendrite Complexity** *Cell Rep* **15**:1037–1050 https://doi.org/10.1016/j.celrep.2016.03.093

Phillips G. R., Tanaka H., Frank M., Elste A., Fidler L., Benson D. L., Colman D. R. (2003) **Gamma-protocadherins are targeted to subsets of synapses and intracellular organelles in neurons** *J Neurosci* **23**:5096–104 https://doi.org/10.1523/JNEUROSCI.23-12-05096.2003

Rawson R. L., Martin E. A., Williams M. E. (2017) **Mechanisms of input and output synaptic specificity: finding partners, building synapses, and fine-tuning communication** *Curr Opin Neurobiol* **45**:39–44 https://doi.org/10.1016/j.conb.2017.03.006

Rubinstein R., Thu C. A., Goodman K. M., Wolcott H. N., Bahna F., Mannepalli S., Ahlsen G., Chevee M., Halim A., Clausen H., Maniatis T., Shapiro L., Honig B. (2015) **Molecular logic of neuronal self-recognition through protocadherin domain interactions** *Cell* **163**:629–42 https://doi.org/10.1016/j.cell.2015.09.026

Serizawa S., Miyamichi K., Takeuchi H., Yamagishi Y., Suzuki M., Sakano H. (2006) **A neuronal identity code for the odorant receptor-specific and activity-dependent axon sorting** *Cell* **127**:1057–69 https://doi.org/10.1016/j.cell.2006.10.031

Steffen D. M., Ferri S. L., Marcucci C. G., Blocklinger K. L., Molumby M. J., Abel T., Weiner J. A. (2021) **The gamma-Protocadherins Interact Physically and Functionally with Neuroligin-2 to Negatively Regulate Inhibitory Synapse Density and Are Required for Normal Social Interaction** *Mol Neurobiol* **58**:2574–2589 https://doi.org/10.1007/s12035-020-02263-z

Suo L., Lu H., Ying G., Capecchi M. R., Wu Q. (2012) **Protocadherin clusters and cell adhesion kinase regulate dendrite complexity through Rho GTPase** *J Mol Cell Biol* **4**:362–76 https://doi.org/10.1093/jmcb/mjs034

Sytnyk V., Leshchyns'Ka I., Schachner M. (2017) **Neural Cell Adhesion Molecules of the Immunoglobulin Superfamily Regulate Synapse Formation, Maintenance, and Function** *Trends Neurosci* **40**:295–308 https://doi.org/10.1016/j.tins.2017.03.003

Tan L., Zhang K. X., Pecot M. Y., Nagarkar-Jaiswal S., Lee P. T., Takemura S. Y., Mcewen J. M., Nern A., Xu S., Tadros W., Chen Z., Zinn K., Bellen H. J., Morey M., Zipursky S. L. (2015) **Ig Superfamily Ligand and Receptor Pairs Expressed in Synaptic Partners in Drosophila** *Cell* **163**:1756–69 https://doi.org/10.1016/j.cell.2015.11.021

Tarusawa E., Sanbo M., Okayama A., Miyashita T., Kitsukawa T., Hirayama T., Hirabayashi T., Hasegawa S., Kaneko R., Toyoda S., Kobayashi T., Kato-Itoh M., Nakauchi H., Hirabayashi M., Yagi T., Yoshimura Y. (2016) **Establishment of high reciprocal connectivity between clonal cortical neurons is regulated by the Dnmt3b DNA methyltransferase and clustered protocadherins** *BMC Biol* **14** https://doi.org/10.1186/s12915-016-0326-6

Toyoda S., Kawaguchi M., Kobayashi T., Tarusawa E., Toyama T., Okano M., Oda M., Nakauchi H., Yoshimura Y., Sanbo M., Hirabayashi M., Hirayama T., Hirabayashi T., Yagi T. (2014) **Developmental epigenetic modification regulates stochastic expression of clustered protocadherin genes, generating single neuron diversity** *Neuron* **82**:94–108 https://doi.org/10.1016/j.neuron.2014.02.005

Wada T., Wallerich S., Becskei A. (2018) **Stochastic Gene Choice during Cellular Differentiation** *Cell Rep* **24**:3503–3512 https://doi.org/10.1016/j.celrep.2018.08.074

Wang X., Weiner J. A., Levi S., Craig A. M., Bradley A., Sanes J. R. (2002) **Gamma protocadherins are required for survival of spinal interneurons** *Neuron* **36**:843–54 https://doi.org/10.1016/s0896-6273(02)01090-5

Weiner J. A., Wang X., Tapia J. C., Sanes J. R. (2005) **Gamma protocadherins are required for synaptic development in the spinal cord** *Proc Natl Acad Sci U S A* **102**:8–14 https://doi.org/10.1073/pnas.0407931101

Wu Q., Maniatis T. (1999) **A striking organization of a large family of human neural cadherin-like cell adhesion genes** *Cell* **97**:779–90 https://doi.org/10.1016/s0092-8674(00)80789-8

Yagi T. (2012) **Molecular codes for neuronal individuality and cell assembly in the brain** *Front Mol Neurosci* **5** https://doi.org/10.3389/fnmol.2012.00045

Yogev S., Shen K. (2014) **Cellular and molecular mechanisms of synaptic specificity** *Annu Rev Cell Dev Biol* **30**:417–37 https://doi.org/10.1146/annurev-cellbio-100913-012953

Author information

Yi-Jun Zhu

Institute of Neuroscience and State Key Laboratory of Neuroscience, CAS Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, 200031, China, Lingang Laboratory, Shanghai Center for Brain Science and Brain-Inspired Intelligence Technology, Shanghai 201210, China, University of Chinese Academy of Sciences, Beijing 100049, China

Cai-Yun Deng

Institute of Neuroscience and State Key Laboratory of Neuroscience, CAS Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, 200031, China

Liu Fan

Lingang Laboratory, Shanghai Center for Brain Science and Brain-Inspired Intelligence Technology, Shanghai 201210, China

Ya-Qian Wang

Lingang Laboratory, Shanghai Center for Brain Science and Brain-Inspired Intelligence Technology, Shanghai 201210, China

Hui Zhou

Lingang Laboratory, Shanghai Center for Brain Science and Brain-Inspired Intelligence Technology, Shanghai 201210, China

Hua-Tai Xu
Lingang Laboratory, Shanghai Center for Brain Science and Brain-Inspired Intelligence Technology, Shanghai 201210, China
For correspondence: xuht@lglab.ac.cn
ORCID iD: 0000-0002-1113-0455

Editors

Reviewing Editor
Kang Shen
Stanford University, United States of America

Senior Editor
Lu Chen
Stanford University, United States of America

Reviewer #1 (Public Review):

The manuscript by Zhu and colleagues aimed to clarify the importance of isoform diversity of PCDHg in establishing cortical synapse specificity. The authors optimized 5' single-cell sequencing to detect cPCDHg isoforms and showed that the pyramidal cells express distinct combinations of PCDHg isoforms. Then, the authors conducted patch-clamp recordings from cortical neurons whose PCDHg diversity was disrupted. In the elegant experiment in Figure 3, the authors demonstrated that the neurons expressing the same sets of cPCDHg isoforms are less likely to form synapses with each other, suggesting that identical cPCDHg isoforms may have a repulsive effect on synapse formation. Importantly, this phenomenon was dependent on the similarity of the isoforms present in neurons but not on the amount of proteins expressed.

One of the major concerns in an earlier version was whether PCDHg isoforms, which are expressed at a much lower level than C-type isoforms, have true physiological significance. The authors conducted additional experiments to address this point by using PCDHg cKO and provided convincing data supporting their conclusion. The results from PCDHg C4 overexpression, showing no impact on synaptic connectivity, further clarified the importance of isoforms. I have no further concerns, however, I would like to point out that the evidence for the necessity of the PCDHg isoform is still lacking because most experiments were done by overexpression. It would be helpful for the readers if the authors could add this point to the discussion.

Reviewer #2 (Public Review):

This short manuscript by Zhu et al. describes an investigation into the role of gamma protocadherins in synaptic connectivity in the mouse cerebral cortex. First, the authors conduct a single-cell RNA-seq survey of postnatal day 11 mouse cortical neurons, using an adapted 10X Genomics method to capture the 5' sequences that are necessary to identify individual gamma protocadherin isoforms (all 22 transcripts share the same three 3' "constant" exons, so standard 3'-biased methods can't distinguish them). This method adaptation is an advance for examining individual gamma transcripts, and it is helpful to publish the method, the characterization of which is improved in this revised manuscript. The results largely confirm what was known from other approaches, which is that a few of the 19 A and B subtype gamma protocadherins are expressed in an apparently stochastic and combinatorial fashion in each cortical neuron, while the 3 C subtype genes are expressed ubiquitously. Second, using elegant paired electrophysiological recordings, the authors show that in gamma protocadherin cortical slices, the likelihood of two neurons on layers 2/3 being synaptically connected is increased. That suggests that gamma protocadherins generally inhibit synaptic connectivity in the cortex; again, this has been reported previously using morphological assays, but it is important to see it confirmed here with physiology. Finally, the authors use an impressive sequential in utero electroporation method to provide evidence that the degree of isoform matching between two neurons negatively regulates their reciprocal synaptic connectivity. These are difficult experiments to do, and while some caveats remain, the main result is consistent. Strengths include the impressive methodology and improved demonstration of the previously-reported finding that gamma protocadherins work via homophilic matching to put a brake on synapse formation in the cortex. Weaknesses include the writing, which even in the revision fails to completely put the new results in context with prior work, which together has largely shown similar results; a still-incomplete characterization of a new alpha protocadherin KO mouse (a minor point but it should still be addressed); and a lack of demonstration of protein levels in electroporated brains. Because of the unique organization and expression pattern of the gamma protocadherins, it is unlikely that these results will be directly applicable to the broader understanding of the role of cell adhesion molecules in synapse development. However, the methodology, which is now better described, should be applicable more broadly and the improved demonstration of the role of gamma protocadherin's negative role in cortical synaptogenesis is helpful.

Reviewer #3 (Public Review):

In this study, Zhu and authors investigate the expression and function of the clustered Protocadherins (cPcdhs) in synaptic connectivity in the mouse cortex. The cPcdhs encode a large family of cadherin-related transmembrane molecules hypothesized to regulate synaptic specificity through combinatorial expression and homophilic binding between neurons expressing matching cPcdh isoforms. But the evidence for combinatorial expression has been limited to a few cell types, and causal functions between cPcdh diversity and wiring specificity have been difficult to test experimentally. This study addresses two important but technically challenging questions in the mouse cortex: 1) Do single neurons in the cortex express different cPcdh isoform combinations? and 2) Does Pcdh isoform diversity or particular combinations among pyramidal neurons influence their connectivity patterns? Focusing on the Pcdh-gamma subcluster of 22 isoforms, the group performed 5'end-directed single-cell RNA sequencing from dissociated postnatal (P11) cortex. To address the functional role of Pcdhg diversity in cortical connectivity, they asked whether the Pcdhgs and isoform matching influence the likelihood of synaptic pairing between 2 nearby pyramidal neurons. They performed simultaneous whole-cell recordings of 6 pyramidal neurons in cortical slices, and measured paired connections by evoked monosynaptic responses. In these experiments, they measured synaptic connectivity between pyramidal neurons lacking the Pcdhgs, or overexpressing dissimilar or matching sets of Pcdhg isoforms introduced by electroporation of plasmids encoding Pcdhg cDNAs.

Overall, the study applies elegant methods that demonstrate that single cortical neurons express different combinations of Pcdh-gamma isoforms, including the upper layer Pyramidal cells that are assayed in paired recordings. The electrophysiology data demonstrate that nearby Pyramidal neurons lacking the entire Pcdhg cluster are more likely to be synaptically connected compared to the control neurons, and that overexpression of matching isoforms between pairs decreases the likelihood to be synaptically connected. These are important and compelling findings that advance the idea that the Pcdhgs are important for cortical synaptic connectivity, and that the repertoire of isoforms expressed by neurons influence their connectivity patterns potentially through a self/non-self discrimination mechanism. However, the findings are limited to probability in connectivity and do they do not support the authors' conclusions that Pcdhg isoforms regulate synaptic specificity, 'by preventing synapse formation with specific cells' or to 'unwanted partners'. Characterizations of the cellular basis of these defects are needed to determine whether they are secondary to other roles in cell positioning, axon/dendrite branching and synaptic pruning, and overall synaptic formation. Claims that Pcdh-alpha and Pcdhg C-type isoforms are not functionally required are premature, due to limitations of the experiments. Moreover, claims that 'similarity level of γ-PCDH isoforms between neurons regulate the synaptic formation' are not supported due to weak statistical analyses presented in Fig4. The overstatements should be corrected. There was also missed opportunity to clearly discuss these results in the context of other published work, including recent publications focused on the cortex.

Strengths:

- The 5' end sequencing with a Pcdhg-amplified library is a technical feat and addresses the pitfall of conventional scRNA-Seq methods due to the identical 3'sequences shared by all Pcdhg isoform and the low abundance of the variable exons. New figures with annotated cell types confirm that several pyramidal and inhibitory cortical subpopulations were captured.

-Statistical assessment of co-occurrence of isoform expression within clusters is also a strength.

- By establishing the combinatorial expression of Pcdhgs by maturing pyramidal cells, the study further substantiates the 'single neuron combinatorial code for cPcdhs' model. Although combinatorial expression is not universal (ie. serotonergic neurons), there was limited evidence. The findings that individual pyramidal neurons express ~1-3 variable Pcdhg transcripts plus the C-type transcripts aligns with single RT-PCR studies of single Purkinje cells (Esumi et al 2005; Toyoda et al 2014). They differ from the findings by Lv et al 2022, where C-type expression was lower among pyramidal neurons. OSNs also do not substantially express C-type isoforms (Mountoufaris et al 2017; Kiefer et al 2023). Differences, and the advantages of the 5'end -directed sequencing (vs. SmartSeq) could be raised in the discussion.

- Simultaneous whole-cell recordings and pairwise comparisons of pyramidal neurons is a technically outstanding approach. They assess the effects of Pcdhg OE isoform on the probability of paired connections.

- The connectivity assay between nearby pairs proved to be sensitive to quantify differences in probability in Pcdhg-cKO and overexpression mutants. The comparisons of connectivity across vertical vs lateral arrangement are also strengths. Overexpressing identical Pcdhg isoform (whether 1 or 6) reduces the probability of connectivity, but there are caveats to the interpretations (see below).

Weaknesses:

-The experiments support a role for the Pcdhgs in influencing the probability of synaptic connectivity between nearby pairs but are not sufficient evidence for synapse specificity. The cPcdhs play multiple roles in neurite arborization, synaptic density, and cell positioning. Kostadinov 2015 also showed that starburst cells lacking the Pcdhgs maintained increased % connectivity at maturity, suggesting a lack of refinement in the absence of Pcdhgs. The known roles raise questions on how these manipulations might have primary effects in these processes and then subsequently impact the probability of connectivity. Investigations of morphological aspects of pyramidal development would strengthen the study and potentially refine the findings. The authors should more clearly relate their findings to the body of cPcdh studies in the discussion.

- Pcdhg cKO-dependent effects on connectivity occur between closely spaced soma (50-100um - Figure 2E), highlighting the importance of spatial arrangement to connectivity (also noted by Tarusawa 2016). Was distance considered for the overexpression (OE) assays, and did the authors note changes in cell distribution which might diminish the connectivity? Recent work by Lv et al 2022 reported that manipulating Pcdhgs influences the dispersion of clonally-related pyramidal neurons, which also impacts the likelihood of connections. Overexpression of Pcdhgc3 increased cell dispersion and decreased the rate of connectivity between pairs. Though these papers are mentioned, they should be discussed in more detail and related to this work.

- Though the authors added suggested citations and improved the contextualization of the study, several statements do not accurately represent the cited literature. It is at the expense of crystalizing the novelty and importance of this present work. For instance, Garrett et al 2012 PMID: 22542181 was the first to describe roles for Pcdhgs in cortical pyramidal cells and dendrite arborization, and that pyramidal cell migration and survival are intact. Line 52 cited Wang et al 2002, but this was limited to gross inspection. Garrett et al is the correct citation for: 'The absence of γ-PCDH does not cause general abnormality in the development of the cerebral cortex, such as cell differentiation, migration, and survival (Wang et al., 2002).' Second, single cell cPcdh diversity is introduced very generally, as though all neuron types are expected to show combinatorial variable expression with ubiquitous C-Type expression. But those initial studies were limited to Purkinje cells (Esumi 2005 and Toyoda 2014). Profiling of serotonergic neurons and OSN reveals different patterns (citations needed for Chen 2017 PMID: 28450636; Mountofaris et al PMID: 2845063; Canzio 2023 PMID: 37347873), raising the idea that cPcdh diversity and ubiquitous C-type expression is not universal. Thus, the authors missed the opportunity to emphasize the gap regarding cPcdh diversity in the cortex.

- They have not shown rigorously and statistically that the rate of connectivity changes with% isoform matching. In Figure 4D, comparisons of % isoform matching in OE assays show a single statistical comparison between the control and 100% groups, but not between the 0%, 11% and 33% groups. Is there a significant difference between the other groups? Significant differences are claimed in the results section, but statistical tests are not provided. The regression analysis in 4E suggests a correlation between % isoform similarity and connectivity probability, but this is not sound as it is based on a mere 4 data points from 4D. The authors previously explained that they cannot evaluate the variance in these recordings as they must pool data together. However, there should be some treatment of variability, especially given the low baseline rate of connectivity. Or at the very least, they should acknowledge the limitations that prevent them from assessing this relationship. Claims in lines 230+ are not supported: ' Overall, our findings demonstrate a negative correlation between the probability of forming synaptic connections and the similarity level of γ-PCDH isoforms expressed in neuron pairs (Fig. 4E)".

-Figure 4 provides connectivity probability, but this result might be affected by overall synapse density. Did connection probability change with directionality (e.g between red to green cells, or green to red cells).

-Generally, the statistical approaches were not sufficiently described in the methods nor in the figure legends, making it difficult to assess the findings. They do not report on how they calculated FDR for connectivity data, when this is typically used for larger multivariate datasets.

- The possibility that the OE effects are driven by total Pcdhg levels, rather isoform matching, should be examined. As shown by qRT-PCR in Fig. 3, expression of individual isoforms can vary. It is reasonable that protein levels cannot be measured by IHC, although epitope tags could be considered as C-terminal tagging of cPcdhs preserves the function in mice (see Lefebvre 2008). Quantification of constant Pcdhg RNA levels by qRT-PCR or sc-RT-PCR would directly address the potential caveat that OE levels vary with isoform combinations.

-A caveat for the relative plasmid expression quantifications in Figure 3-S1 is that IHC was used to amplify the RFP-tagged isoform, and thus does not likely preserve the relationship between quantities and detection.

-Figure 1 didn't change in response to reviews to improve clarity. New panels relating to the scRNA-Seq analyses were added to supplementary data but many are central and should be included in Figure 1 (ie. S1-Fig6D). In the Results, the authors state that neuronal subpopulations generally show a combinatorial expression of some variable RNA isoforms and near ubiquitous C-type expression. But they only show data for the Layer 2/3 neuron-specific cluster in S1-Fig-6D, and so it is not clear if this pattern applies to other clusters. Fig. S1-5 show a low number of expressed isoforms per cell, but specific descriptions on whether these include C-type isoforms would be helpful. Figure 1F showing isoform profile in all neurons is not particularly meaningful. There is a lot of interest in neuron-type specific differences in cPcdh diversity, and the authors could highlight their data from S1-5 accordingly.

-The concept of co-occurrence and results should be explained within the results section, to more clearly relate this concept to data and interpretations. Explanations are now found in the methods, but this did not improve the clarity of this otherwise very interesting aspect of the study.

- The claim that C-type Pcdhgs do not functionally influence connectivity is premature. Tests were limited to PcdhgC4, which has unique properties compared to the other 2 C-type isoforms (Garrett et al 2019 PMID: 31877124; Mancia et al PMID: 36778455). The text should be corrected to limit the conclusion to PcdhgC4, and not generally to C-type. The authors should test PcdhgC3 and PcdhgC5 isoforms.

-The group generated a novel conditional Pcdh-alpha mouse allele using CRISPR methods, and state that there were no changes in synaptic connectivity in these Pcdh-alpha mutants. But this claim is premature. The Southern blots validate the targeting of the allele. But further validations are required to establish that this floxed allele can be efficiently recombined, disrupting Pcdha protein levels and function. Pcdha alleles have been validated by western blots and by demonstration of the prominent serotonergic axonal phenotype of Pcdha-KO (ie. Chen 2017 PMID: 28450636; Ing-Esteves 2018 PMID: 29439167).

-The Discussion would be strengthened by a deeper discussion of the findings to other cPcdh roles and studies, and of the limitations of the study. The idea that the Pcdhgs are influencing the rate of connectivity through a repulsion mechanism or synaptic formation (ie through negative interactions with synaptic organizers such as Nlgn - Molumby 2018, Steffen 2022) could be presented in a model, and supported by other literature.